Software Requirements Document[1,2]

Requirements form the bridge between business objectives and software implementation. They serve as the foundation for all design, development process, testing, and documentation activities. Clear, testable requirements are crucial to building software that meets stakeholder needs.

The following template is rather "general", in that not every SRD will have all the sections herein, so they could be left out of a given SRD. However, they still might be included for completeness (indicating "not applicable" or some such) so that the reviewer/customer knows you've at least considered them.

1. Introduction and Purpose

A general description of the system under development (SUD) and its intended usage/purpose.

- *The System Under Development (SUD) is a programmed chess application that allows a user to play against an opponent. The opponent can either be a real-life player or an Artificial Intelligence (AI) bot powered by an Application Programming Interface (API). The SUD is intended to allow players to view a fully visualized chess board and pieces and interact with them to compete in a game of chess following standard chess rules.*

1.1 Acronyms, Terms and References

List acronyms, terms and references used in the SRD.

Acronym	Meaning

[1] https://insights.sei.cmu.edu/documents/3021/2006_017_001_22987.pdf
[2] https://www.computer.org/resources/software-requirements-specifications

SRD	Software Requirements Document
SUD	System Under Development
AI	Artificial Intelligence
API	Application Programming Interface

Reference	Source
FIDE Laws of Chess	https://www.fide.com/FIDE/handbook/LawsOfChess.pdf

2. Business Objectives

Define the business objectives driving creation of the SUD. These may or may not be stated formally as requirements. However, at a minimum, defining these objectives provides the basis for everything else.

- *The SUD is being developed for a course project, with the primary objectives including demonstrating software requirements engineering, system design, and rule-based logic implementation.*

3. User Personas or Roles

Define the categories of users and/or the roles they play in the eventual end system.

- *The SUD is intended to be used primarily by college students to play chess following standard chess rules.*
- *Users shall understand the rules of chess as outlined by the FIDE Laws of Chess Articles 1, 2, 3, and 5.*
- *Users shall be able to understand algebraic chess notation as described by the FIDE Laws of Chess Appendix C.*

4. Feature List

Define the high-level features of the SUD.

- *[SRD_FL_01]: The SUD shall display a chess board with the current game state.*
- *[SRD_FL_02]: The SUD shall accept user input to update the game state.*
- *[SRD_FL_03]: The SUD shall determine if the user has entered a legal move according to the FIDE Laws of Chess.*
- *[SRD_FL_04]: The SUD shall request AI move generation from a chess engine API.*

5. Use Cases[3]

Define the allowed/typical user interactions with the eventual end system, including success & failure cases, exceptions, etc.

- *UC_01: Start New Game*
    - *Actor: User*
    - *Description: The user initiates a new chess game. The system initializes the game environment by resetting the board state, placing all pieces in their standard starting positions, and preparing the system for opponent selection and gameplay.*
    - *Preconditions: The SUD is running. No active chess game is in progress, or the user has elected to start a new game.*
    - *Normal Flow:*
        1. *The user selects the option to start a new game.*
        2. *The SUD clears any existing game state.*
        3. *The SUD initializes a new chess board.*
        4. *The SUD places all chess pieces in their standard starting positions.*
        5. *The SUD prepares the system for opponent selection or gameplay configuration.*
    - *Alternate Flows:*

[3] https://en.wikipedia.org/wiki/Use_case

        - *If a game is already in progress, the SUD prompts the user to confirm restarting the game.*
        - *If the user cancels the action, the existing game state remains unchanged.*
    - *Postconditions: A new game state is created.*
- *UC_02: Select Opponent Type*
    - *Actor: User*
    - *Description: The user selects the type of opponent for a new chess game. The system presents available opponent options, accepts the user's selection, and configures the game mode accordingly before allowing the gameplay to begin.*
    - *Preconditions: The SUD is running, and a new game has not yet started.*
    - *Normal Flow:*
        1. *The user selects the option to start a new game.*
        2. *The SUD displays available opponent options (human or AI).*
        3. *The user selects the desired opponent type.*
        4. *The SUD records the selected opponent type.*
        5. *The SUD configures the game mode based on the selection.*
        6. *The SUD initializes the game.*
    - *Alternate Flows: The SUD prompts the user to make a selection if no opponent type is selected. If the user exits the selection process, the game does not start.*
    - *Postconditions: The opponent type is defined, and the system is ready to initialize gameplay.*

6. User Requirements[4]

Define end-user expectations as requirements.

- *[SRD_UR_01]: The SUD shall enable users to start a new game of chess.*

[4] https://perfval.com/technical-discussion/user-requirements/

- *[SRD_UR_02]: The SUD shall enable users to choose between a human or AI opponent.*
- *[SRD_UR_03]: The SUD shall prevent users from making illegal chess moves.*
- *[SRD_UR_04]: The SUD shall provide feedback to the user when an invalid move is attempted.*
- *[SRD_UR_05]: The SUD shall inform users when the game ends.*
- *[SRD_UR_06]: The SUD shall inform users of the reason the game was terminated.*
- *[SRD_UR_07]: The SUD shall support single player vs AI play.*
- *[SRD_UR_08]: The SUD shall support local two player chess play.*

7. Functional Requirements[5]

Define the functional requirements of the SUD.

- *[SRD_FR_01]: The SUD shall initialize all pieces as outlined in FIDE Laws of Chess Article 2.2.*
- *[SRD_FR_02]: The SUD shall determine legal pawn moves in compliance with FIDE Laws of Chess 3.1 and 3.7.*
- *[SRD_FR_03]: The SUD shall determine legal rook moves in compliance with FIDE Laws of Chess 3.1, 3.3, and 3.5.*
- *[SRD_FR_04]: The SUD shall determine legal knight moves in compliance with FIDE 3.1 and 3.6.*
- *[SRD_FR_05]: The SUD shall determine legal bishop moves in compliance with FIDE 3.1, 3.2, and 3.5.*
- *[SRD_FR_06]: The SUD shall determine legal queen moves in compliance with FIDE 3.1, 3.4, and 3.5.*
- *[SRD_FR_07]: The SUD shall determine legal king moves in compliance with FIDE 3.1, 3.8, and 3.9.*
- *[SRD_FR_08]: The SUD shall give the user the option to play against an AI player.*

[5] https://en.wikipedia.org/wiki/Functional_requirement

- *[SRD_FR_09]: The SUD shall give the user the option to play against a real-life player.*

8. Nonfunctional Requirements[6]

Define the set of nonfunctional requirements of the SUD.

- *[SRD_NFR_01]: The SUD shall comply with the rules of chess as listed by the FIDE Laws of Chess Article 1.*
- *[SRD_NFR_02]: The SUD shall comply with the rules of chess as listed by the FIDE Laws of Chess Article 2.*
- *[SRD_NFR_03]: The SUD shall comply with the rules of chess as listed by the FIDE Laws of Chess Article 3.*
- *[SRD_NFR_04]: The SUD shall comply with the rules of chess as listed by the FIDE Laws of Chess Article 5.*

9. Interface Requirements[7]

Define the set of interface requirements of the SUD.

- *[SRD_IR_01]: The SUD shall use an API to access a chess engine for AI move generation.*
- *[SRD_IR_02]: The SUD shall provide a graphical user interface for piece selection and movement.*
- *[SRD_IR_03]: The SUD shall provide an interface option to start a new game.*
- *[SRD_IR_04]: The SUD shall provide an interface option to select the opponent type.*
- *[SRD_IR_05]: The SUD shall provide an interface option to resign from an active game.*

[6] https://en.wikipedia.org/wiki/Non-functional_requirement
[7] https://argondigital.com/resource/whitepapers/everything-you-wanted-to-know-about-interface-requirements/

- *[SRD_IR_06]: The SUD shall send the current game state to the chess engine API when an AI move is required.*

10. Performance Requirements[8]

Define the set of performance requirements of the SUD.

- *[SRD_PR_01]: The SUD shall take no longer than 50 milliseconds to process user inputs and update the game state accordingly.*
- *[SRD_PR_02]: The SUD shall take no longer than 1 second to calculate the chess engine moves and update the game state accordingly.*

11. Availability Requirements[9]

Define the set of availability requirements of the SUD.

- *[SRD_AR_01]: The SUD shall notify the user if it cannot access the chess engine API.*
- *[SRD_AR_02]: The SUD shall not allow the user to play against the AI if it cannot access the chess engine API.*
- *[SRD_AR_03]: The SUD shall be available whenever the host system is operational.*

12. Safety Requirements[10]

Define the set of safety requirements of the SUD.

- *It is not necessary for the software to have safety requirements as it does not have any chance to cause harm to humans or other systems.*

[8] https://www.ibm.com/docs/en/aix/7.2?topic=implementation-performance-requirements-documentation
[9] https://requirements.com/Content/What-is/what-are-availability-requirements
[10] *Need a good reference here.*

13. Security Requirements[11]

Define the set of security requirements of the SUD.

- *It is not necessary for the software to have security requirements as it can not cause any harm or damage to humans or other systems.*

14. Design and Implementation Constraints[12]

Define any design & implementation constraints placed on the SUD.

- *[SRD_DIC_01]: The SUD shall be implemented using the Python programming language.*

15. External System Requirements[13]

Define the set of external system requirements of the SUD.

- *[SRD_ESR_01]: The SUD shall communicate with an external chess engine API to generate AI moves.*

16. Quality Assurance Requirements[14]

Define the set of quality assurance requirements of the SUD.

- *[SRD_QAR_01]: The SUD shall allow completion of a full chess game from initial setup to checkmate, stalemate, resignation, or draw without system failure.*
- *[SRD_QAR_02]: The SUD shall correctly detect checkmate, stalemate, and draw conditions according to FIDE Article 5.*

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[11] https://requirements.com/Content/What-is/what-are-security-requirements-2
[12] Need a good reference here.
[13] Need a good reference here.
[14] https://soma.larc.nasa.gov/stp/dynamic/pdf_files/NASA-STD-87398B%20SoftwareAssuranceSafety.pdf

- *[SRD_QAR_03]: Functional requirements of the SUD shall be verified through test cases.*

17. Documentation Requirements

Define the list of documents required with delivery of the SUD.

- *[SRD_DR_01]: The SUD shall have trace tags on every requirement, design, code, and test for the SUD.*
- *[SRD_DR_02]: The SUD shall have an SRD describing the requirements of the system.*

18. Trace Tags

List the trace tags defined in the SRD.

- *SRD_FL: Feature List*
- *SRD_UC: Use Case*
- *SRD_UR: User Requirement*
- *SRD_FR: Functional Requirement*
- *SRD_NFR: Nonfunctional Requirement*
- *SRD_IR: Interface Requirement*
- *SRD_PR: Performance Requirement*
- *SRD_AR: Availability Requirement*
- *SRD_DIC: Design and Implementation Constraint*
- *SRD_ESR: External System Requirement*
- *SRD_QAR: Quality Assurance Requirement*
- *SRD_DR: Documentation Requirement*

Appendix

Requirements Analysis & Review[15]

Requirements analysis is the process of examining requirements to improve quality, uncover issues, and gain a deeper understanding of what stakeholders want and need. Why do we analyze requirements? It serves several important purposes:

- *Validate completeness – Ensure all necessary capabilities are captured.*
- *Remove ambiguity – Refine vague, unclear requirements.*
- *Check consistency – Identify conflicting or contradictory requirements.*
- *Assess feasibility – Determine if requirements are realistic and achievable.*
- *Prioritize implementation – Understand relative importance and urgency of each requirement.*
- *Estimate effort – Gauge work needed to satisfy requirements.*
- *Guide design – Derive technical implications and decisions from requirements.*

Analyzing requirements reduces risk by catching problems early when they are easiest to fix. It also sets the stage for developers to build the right solution.

[15] See the companion sample SRD review checklist.